

INTERNSHIP PROJECT REPORT

Smart Battery Management System

ESP32 Devkit C v4 · Wokwi Simulator · Blynk Cloud IoT

1. Introduction

This report documents the design, implementation, and validation of a Smart Battery Management System (BMS) developed as part of the ElevanceSkills Technology internship programme. The project spans six interconnected embedded and IoT tasks constituting a production-grade solution for real-time multi-cell lithium battery monitoring.

The system runs on an ESP32 DevKit C v4, simulated using the Wokwi embedded simulator, and connected to the Blynk Cloud IoT platform for remote monitoring. Four slide potentiometers simulate individual Li-ion cell voltages; a relay module, active buzzer, and I2C LCD provide physical outputs.

Core design principles:

- Non-blocking execution — zero use of delay() anywhere in the main loop
- Modular task separation — each of the six tasks is a self-contained subsystem
- Production-grade fault tolerance — hardware anomalies isolated without system crash
- Intelligent cloud telemetry — event-driven, not fixed-interval

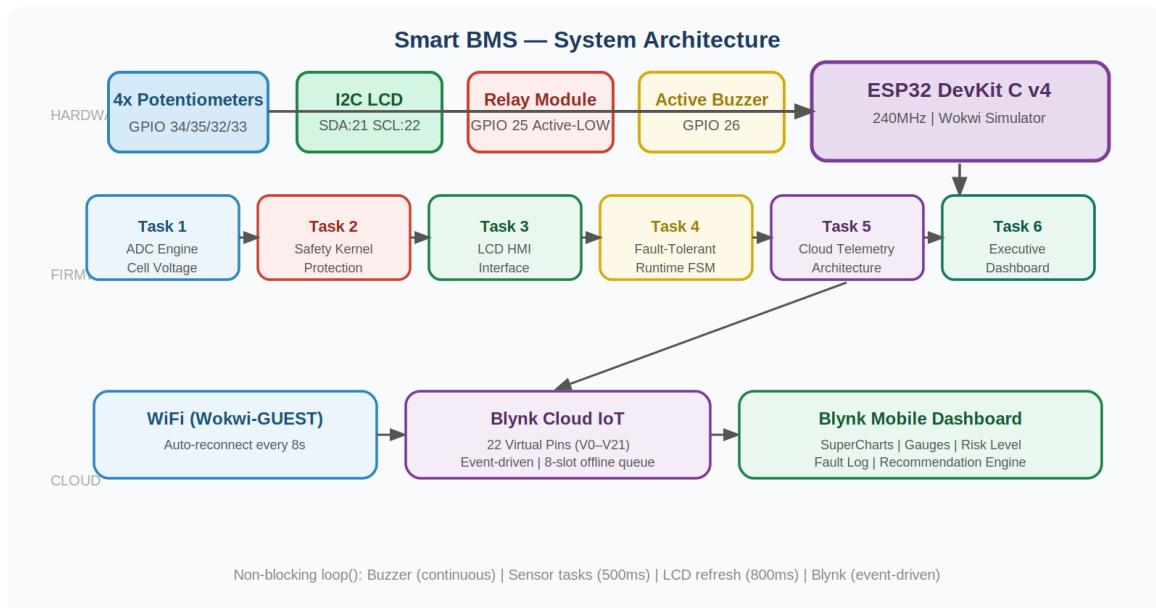


Figure 1 — Smart BMS System Architecture

2. Hardware & Circuit Overview

The table below lists all hardware components, GPIO assignments, and communication protocols.

Component	GPIO / Pin	Protocol	Notes
Slide Pot 1 (Cell 1)	GPIO 34	Analog ADC	8-sample average, maps 3.0–4.5 V
Slide Pot 2 (Cell 2)	GPIO 35	Analog ADC	8-sample average, maps 3.0–4.5 V
Slide Pot 3 (Cell 3)	GPIO 32	Analog ADC	8-sample average, maps 3.0–4.5 V
Slide Pot 4 (Cell 4)	GPIO 33	Analog ADC	8-sample average, maps 3.0–4.5 V
I2C LCD 16×2	SDA: GPIO 21, SCL: GPIO 22	I2C @ 0x27	Differential refresh, 4 rotating screens

Component	GPIO / Pin	Protocol	Notes
Relay Module	GPIO 25	Digital (Active-LOW)	NO contact drives load LED
Active Buzzer	GPIO 26	Digital	300ms ON / 700ms OFF non-blocking

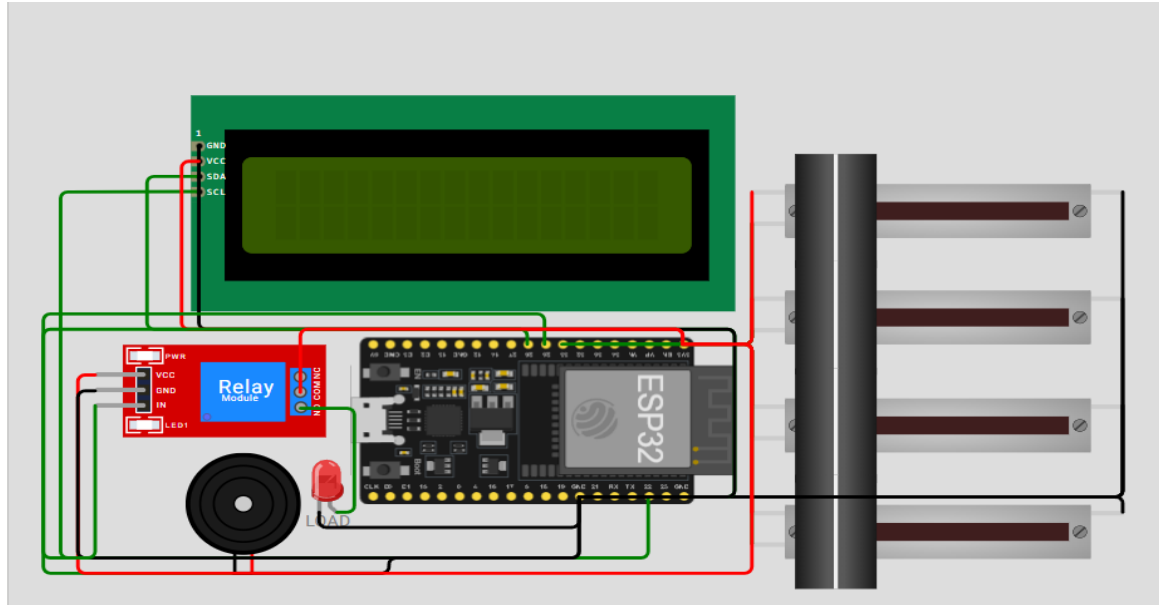


Figure 2 — Wokwi Circuit Diagram (insert screenshot)

3. Tasks Overview

The project comprises six defined tasks, each a self-contained subsystem evaluated independently.

Task	Name	Key Feature	Status
Task 1	Adaptive Multi-Cell Battery Intelligence Engine	8-sample ADC averaging, 4-state health classification	✓ Complete
Task 2	Event-Driven Safety Protection Kernel	4 protection events, relay + buzzer, 6-cycle recovery	✓ Complete
Task 3	Intelligent Embedded HMI & Diagnostic Interface	4 rotating screens, flicker-free differential refresh	✓ Complete
Task 4	Fault-Tolerant Embedded Runtime System	FSM with 4 modes, ring-buffer fault log, frozen ADC detection	✓ Complete
Task 5	Intelligent Cloud Telemetry Architecture	Event-driven Blynk, offline queue, WiFi self-healing	✓ Complete
Task 6	Executive Battery Intelligence Dashboard	22 virtual pins, risk level, 9-scenario recommendation	✓ Complete

4. Task Implementation Details

4.1 Task 1 — Adaptive Multi-Cell Battery Intelligence Engine

Objective: Read and classify four simulated Li-ion cell voltages and compute pack-level analytics in real time.

Each GPIO pin is sampled 8 times and averaged to reduce ADC noise, then linearly mapped from 0–4095 to 3.00–4.50 V. Cells are classified per cycle:

- Faulty: $\text{ADC raw} \leq 3$ or ≥ 4092 for 3 consecutive cycles (open/short circuit simulation)
- Weak: voltage < 3.30 V (below warning threshold V_WARN)
- Overvoltage: voltage > 4.30 V (threshold set 0.20 V below pot max to avoid false triggers)

Pack statistics (average, min, max, imbalance%) are computed from healthy cells only. Four-state health classification:

Health State	Condition	Implication
HEALTHY	Imbalance $< 2\%$, no faults	Normal operation
MINOR IMBALANCE	Imbalance $\geq 2\%$	Schedule balancing
CRITICAL IMBALANCE	Imbalance $\geq 5\%$ or 1 faulty cell	Immediate attention
PACK FAILURE	≥ 2 faulty cells	FAILSAFE mode triggered

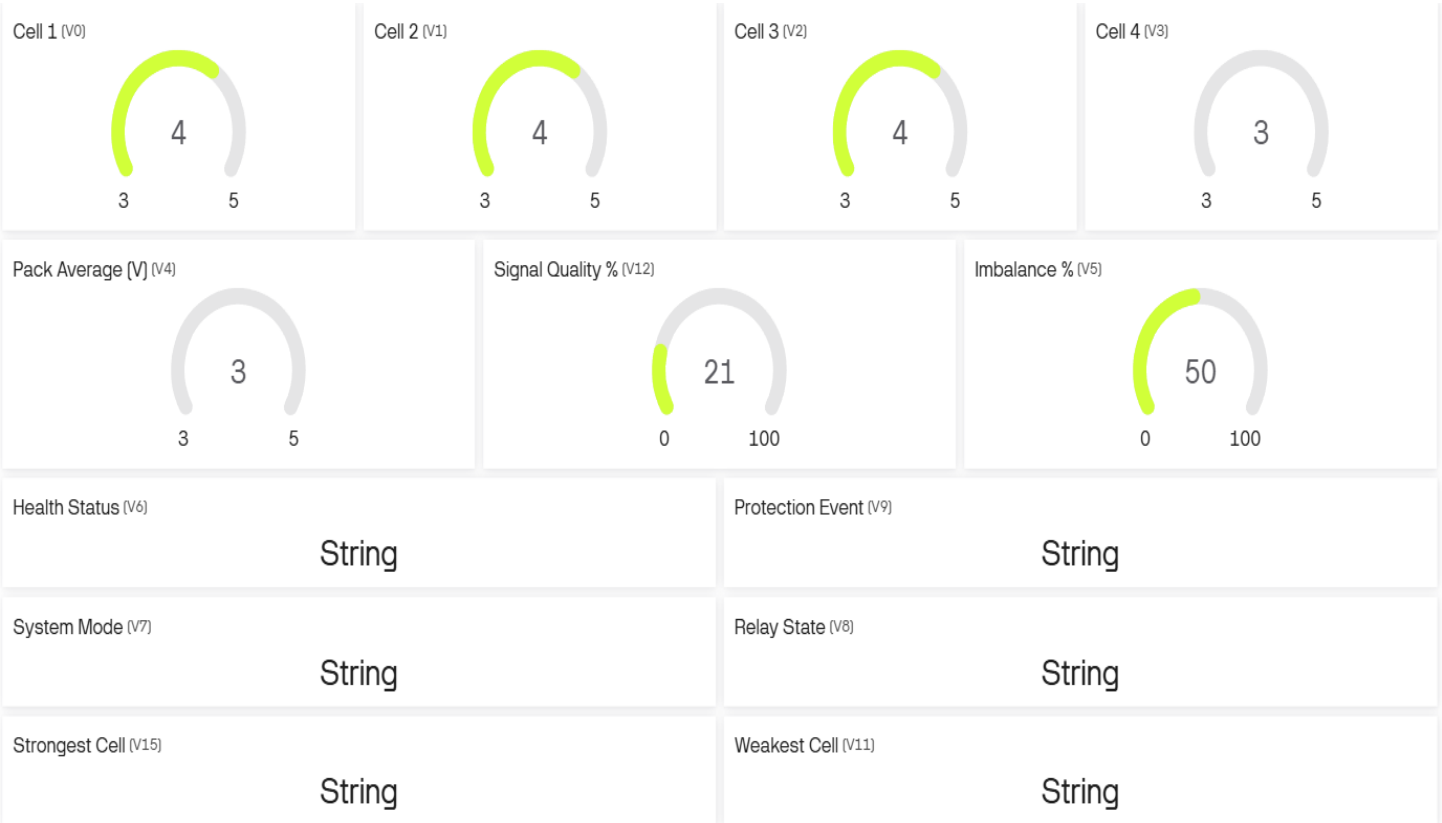


Figure 3 — Dashboard Evidence

4.2 Task 2 — Event-Driven Safety Protection Kernel

Objective: Detect four safety conditions and respond with relay cutoff and buzzer alerts using fully non-blocking millis()-based timing.

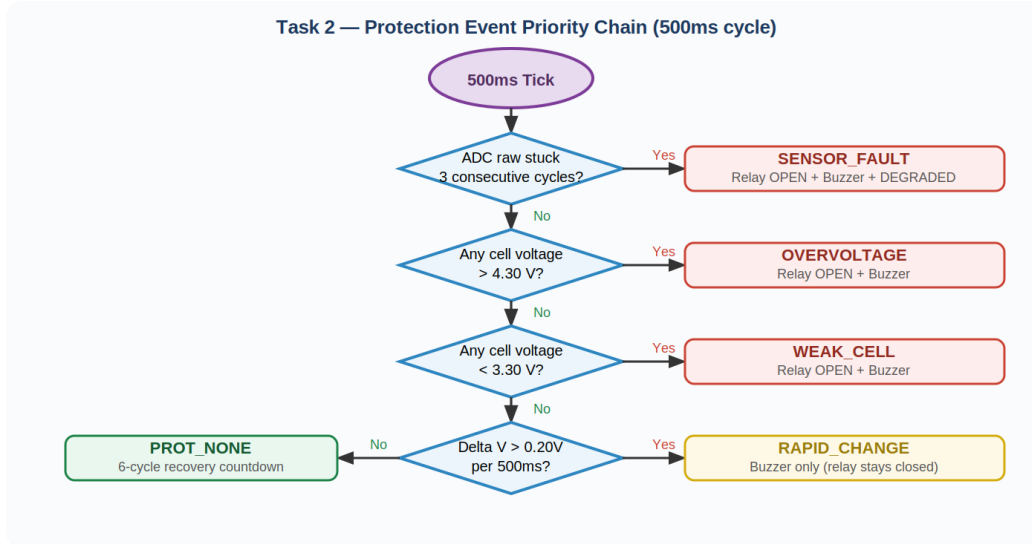


Figure 4 — Protection Event Priority Chain Flowchart

Event	Threshold	Action	Recovery
SENSOR_FAULT	$\text{ADC} \leq 3$ or $\geq 4092 \times 3$ cycles	Relay OPEN + Buzzer + DEGRADED mode	6 clear cycles
OVERVOLTAGE	$> 4.30 \text{ V}$ on any cell	Relay OPEN + Buzzer	6 clear cycles
WEAK_CELL	$< 3.30 \text{ V}$ on any cell	Relay OPEN + Buzzer	6 clear cycles
RAPID_CHANGE	$\Delta V > 0.20 \text{ V}$ per 500ms	Buzzer only (relay stays)	6 clear cycles

Key design: Buzzer timing (300ms ON / 700ms OFF) is checked on every loop() iteration, not inside the 500ms cycle, achieving true non-blocking precision. A 1500ms anti-chatter debounce prevents relay flutter.

4.3 Task 3 — Intelligent Embedded HMI & Diagnostic Interface

Objective: Provide a real-time diagnostic interface on a 16×2 I2C LCD with auto-rotating screens and critical-priority override.

Screen	Content	Rotation
CELLS	Cell voltages C1/C2 → C3/C4 (sub-flip every 2s)	Position 1 of 4
PACK	Avg voltage, imbalance%, weakest/strongest cell	Position 2 of 4
SAFETY	Active protection event, mode, relay state	Position 3 of 4
FAULTS	Total fault count + last fault message	Position 4 of 4
CRITICAL (override)	Active warning label + mode-specific action	Halts rotation

Flicker-free rendering: a differential update compares each 16-char line against previous content and only writes changed lines. Any potentiometer change resets the display immediately to the CELLS screen.

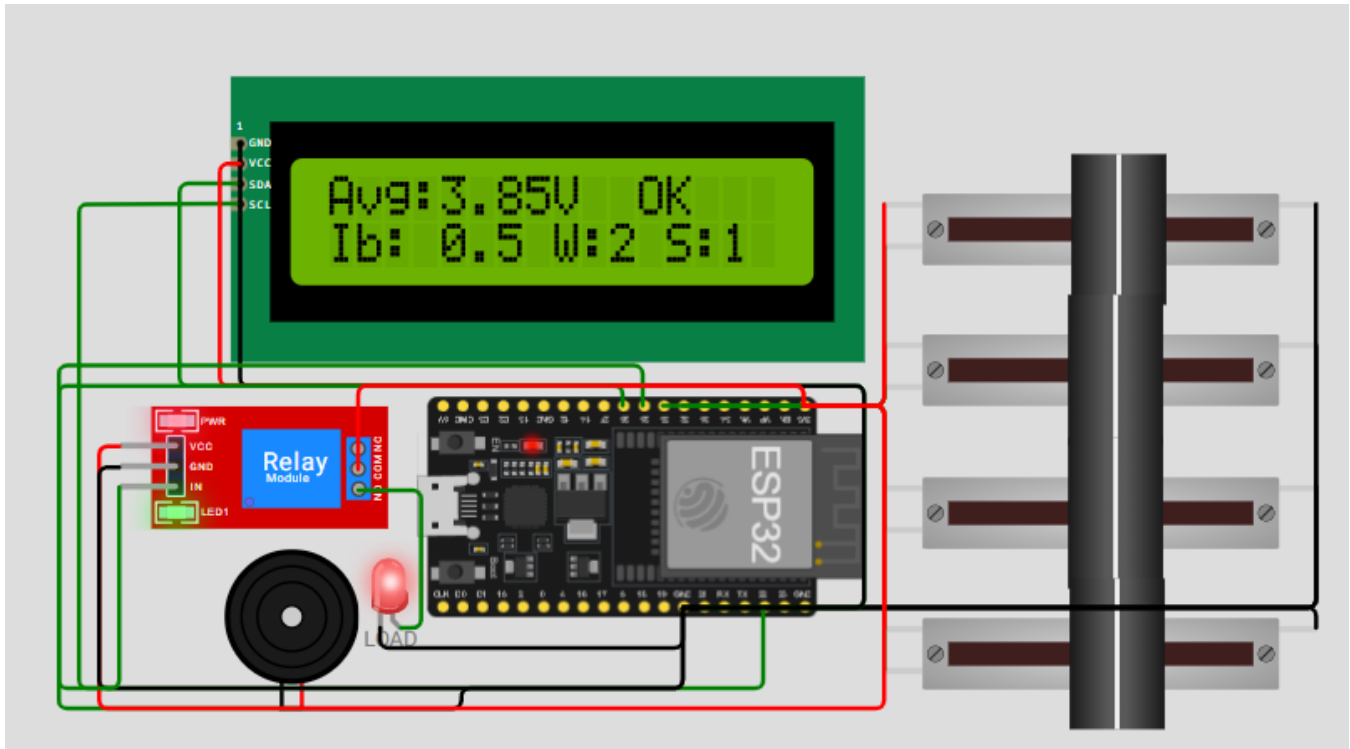


Figure 5 — Task 3 LCD Evidence (insert Wokwi LCD screenshot)

4.4 Task 4 — Fault-Tolerant Embedded Runtime System

Objective: Maintain system integrity through anomaly detection, fault isolation, and a 4-state operational mode FSM with timestamped ring-buffer logging.

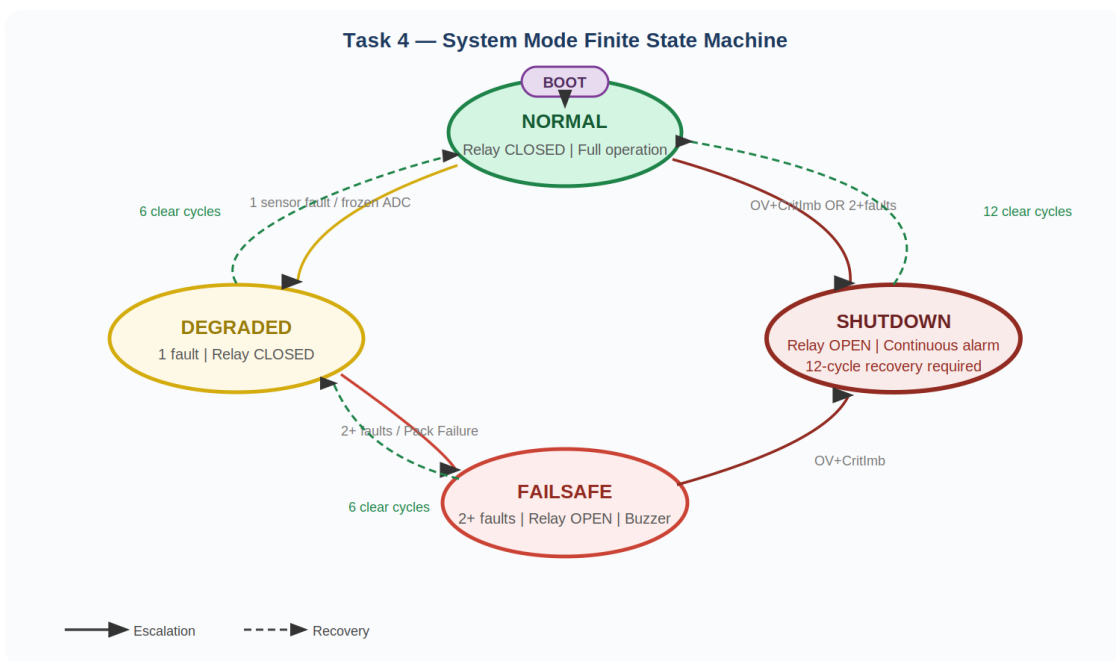


Figure 6 — Task 4 System Mode Finite State Machine

State	Entry Condition	Exit Condition	Behaviour
NORMAL	Boot / all faults cleared	Any fault detected	Full operation, relay closed
DEGRADED	1 sensor fault / frozen ADC / relay mismatch	Fault clears × 6 cycles	Log event, continue
FAILSAFE	≥ 2 faults / Pack Failure	Faults clear × 6 cycles	Relay OPEN, buzzer, halt loads
SHUTDOWN	OV + CritImb OR ≥ 2 faults + Pack Failure	All clear × 12 cycles	Full relay cutoff, continuous alarm

Fault log: 10-entry ring buffer capturing timestamp, message (51 chars), mode-at-fault, and cell voltage snapshot. 5-second deduplication window prevents log flooding.



Figure 7 — Task 4 Dashboard Evidence: Fault Log & Fault Count (insert screenshot)

4.5 Task 5 — Intelligent Cloud Telemetry Architecture

Objective: Implement smart, event-driven IoT telemetry via Blynk Cloud with offline resilience and automatic reconnection replay.

Timer	Interval	Action
Continuous	Every loop()	Buzzer control (300ms ON / 700ms OFF) + Blynk.run()
T_SENSOR	500ms	Read all 4 ADCs, classify cells, compute pack stats, run safety & fault tasks
T_LCD_REFRESH	800ms	Periodic LCD screen refresh
T_LCD_ROTATE	4000ms	Advance LCD screen: CELLS → PACK → SAFETY → FAULTS
T_CELL_FLIP	2000ms	Toggle cell sub-screen: C1/C2 ↔ C3/C4
T_BLYNK_RUN	100ms	Poll Blynk.run() for incoming commands
T_BLYNK_MIN	2000ms	Minimum interval between Blynk data sends
T_WIFI_RETRY	8000ms	WiFi reconnect attempt when disconnected
T_QUEUE_REPLAY	300ms	Replay one queued offline event after reconnection

Offline queue: up to 8 state-change events queued when WiFi or Blynk is unavailable. On reconnection, events are replayed at 300ms intervals before resuming live data. WiFi RSSI is mapped linearly (−100 to −50 dBm → 0–100%) and published to V12.

Signal Quality %

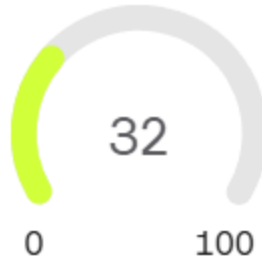


Figure 8 — Task 5 Dashboard Evidence: Signal Quality & Cloud Status (insert screenshot)

4.6 Task 6 — Executive Battery Intelligence Dashboard

Objective: Present a complete, professional IoT monitoring dashboard with risk analysis, operator recommendations, live telemetry, historical trends, and fault history.

Pin	Name	Type	Range	Purpose
V0–V3	Cell 1–4 Voltage	Double	3.0–4.5 V	Live cell voltages from ADC
V4	Pack Average	Double	3.0–4.5 V	Average of healthy cells only
V5	Imbalance %	Double	0–100	Pack imbalance percentage
V6	Health Status	String	—	HEALTHY / MINOR IMB. / CRITICAL / PACK FAILURE
V7	System Mode	String	—	NORMAL / DEGRADED / FAILSAFE / SHUTDOWN
V8	Relay State	String	—	OPEN (Protection ON) / CLOSED (Normal)
V9	Protection State	String	—	Active protection event
V10	Fault Count	Integer	0–9999	Session fault counter
V11	Weakest Cell	String	—	Cell number with lowest voltage
V12	Signal Quality	Integer	0–100%	WiFi RSSI mapped 0–100
V13	Recommendation	String	—	9-scenario operator action string
V14	Risk Level	Integer	0–3	0=Safe, 1=Caution, 2=Warning, 3=Critical
V15	Strongest Cell	String	—	Cell number with highest voltage
V16	Fault Log	String	—	Latest fault entry (timestamp + mode)
V17	Total Faults	Integer	0–9999	Cumulative fault count
V18–V21	Cell 1–4 History	Double	3.0–4.5 V	SuperChart historical trend data

Risk Level mapping: SHUTDOWN / Pack Failure → 3 (CRITICAL); FAILSAFE / Critical Imbalance → 2 (WARNING); DEGRADED / Minor Imbalance → 1 (CAUTION); otherwise → 0 (SAFE). A 9-scenario recommendation engine publishes a specific operator action string to V13 in priority order.

Pin	Name	Data Type	Summary	Min	Max	Default	Units	User Automation
V0	Cell 1 Voltage	Double		0	4.5	3.7	—	10 None
V1	Cell 2 Voltage	Double		0	4.5	3.7	—	10 None
V2	Cell 3 Voltage	Double		0	4.5	3.7	—	10 None
V3	Cell 4 Voltage	Double		0	4.5	3.7	—	10 None
V4	Pack Average	Double		0	4.5	3.7	—	10 None
V5	Imbalance Percent	Double		0	100	0	—	10 None
V6	Health Status	String					—	10 None
V7	System Mode	String					—	10 None
V8	Relay State	String					—	10 None
V9	Protection State	String					—	10 None
V10	Fault Count	Integer		0	9999	0	—	10 None
V11	Weakest Cell	String					—	10 None
V12	Signal Quality	Integer		0	100	0	—	10 None
V13	Recommendation	String					—	10 None
V14	Risk Level	Integer		0	3	0	—	10 None
V15	Strongest Cell	String					—	10 None
V16	Fault Log	String					—	10 None
V17	Total Faults	Integer		0	9999	0	—	10 None
V18	Cell 1 History	Double		0	4.5	3.7	—	10 None
V19	Cell 2 History	Double		0	4.5	3.7	—	10 None
V20	Cell 3 History	Double		0	4.5	3.7	—	10 None
V21	Cell 4 History	Double		0	4.5	3.7	—	10 None

Figure 9 — Blynk Datastreams: All 22 Virtual Pins (V0–V21)

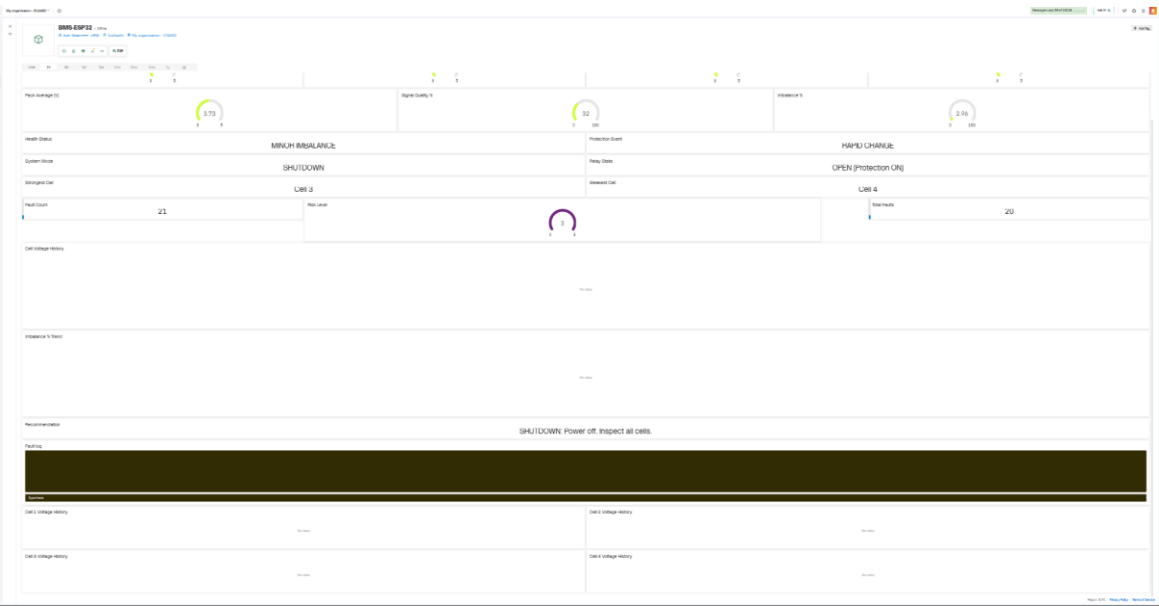


Figure 10 — Blynk Dashboard: SHUTDOWN State (System Mode=SHUTDOWN, Risk Level=3, Relay=OPEN)

Recommendation

HEALTHY: 3.85V Imb:0.5% OK.

Figure 11 — Blynk Dashboard: Healthy State (insert screenshot)

5. Challenges & Solutions

Several non-trivial bugs were identified and resolved across development iterations:

Challenge	Solution Applied
Blynk disconnection due to unthrottled Blynk.run()	Wrapped in 100ms millis() gate (T_BLYNK_RUN) to prevent blocking the main loop
Blynk.config() called repeatedly on every reconnect	Introduced blynkConfigured flag; config called only once; reconnects use Blynk.connect()
Fault history lost when Blynk was offline at fault time	Implemented 8-slot offline event queue with reconnect replay at 300ms intervals
Frozen ADC false positive at boot (rapid change detection)	Added STARTUP_SKIP_CYCLES = 8 to defer rapid-change and frozen-ADC detection
Sticky SHUTDOWN mode — unconditional return in task4_TransitionMode()	Removed unconditional return; SHUTDOWN now recovers via 12-cycle SHUTDOWN_RECOVERY_CYCLES countdown
Buzzer never stopped after SHUTDOWN recovery	Explicit buzzer clear (buzzerActive = false + digitalWrite LOW) added at mode recovery point
Non-atomic relay close on recovery from SHUTDOWN	Added explicit digitalWrite(PIN_RELAY, HIGH) at the moment of mode transition
Relay logic inverted for active-LOW module	Corrected: LOW = OPEN (protection active), HIGH = CLOSED (normal)
Overvoltage false trigger near potentiometer maximum (4.50 V)	Threshold set to 4.30 V instead of 4.20 V, leaving 0.20 V margin below ADC maximum

6. Skills & Competencies

Competency Area	Skills Demonstrated
Embedded C++ / Arduino	ADC sampling, millis()-based non-blocking timers, struct data modelling, ISR-free design
Hardware Interfacing	I2C LCD, relay (active-LOW), buzzer PWM, multi-channel ADC with GPIO configuration
Safety System Design	Priority-chained protection events, debounce, anti-chatter, multi-cycle recovery logic
Finite State Machines	4-state mode FSM with asymmetric recovery, escalation and de-escalation paths
IoT / Cloud Platforms	Blynk Virtual Pins (22), event-driven telemetry, RSSI mapping, offline queue, reconnect replay
Debugging & Testing	Serial dashboard, fault log ring buffer, frozen ADC simulation, startup skip cycle

7. Outcomes & Conclusion

The completed Smart BMS satisfies all six internship task requirements and demonstrates a production-quality embedded IoT system. Key outcomes:

- Fully non-blocking, real-time firmware running six concurrent logical tasks on a single ESP32 core
- A robust safety kernel responding to four distinct fault conditions with relay, buzzer, and cloud alerts
- 4-state FSM with asymmetric recovery that correctly isolates, escalates, and recovers from hardware anomalies
- Professional HMI with four rotating screens, critical-priority override, and flicker-free rendering
- Event-driven IoT telemetry system operating correctly both online and offline, with full reconnect replay
- 22-pin Blynk dashboard: live data, historical trends, fault history, risk level, and operator recommendations

The most significant technical achievements are the fully non-blocking architecture, the priority-chained safety kernel, the fault-tolerant runtime FSM, and the intelligent offline-capable cloud telemetry. Each component was refined through iterative debugging with all issues documented and resolved in the final codebase.

| All 6 Tasks Complete |

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